



Submission Deadline: 15/02/26	Fall 2026	Practice Assignment 2
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6. Use Newton's method to find solutions accurate to within 10^{-5} for the following problems.
- $e^x + 2^{-x} + 2 \cos x - 6 = 0$ for $1 \leq x \leq 2$
 - $\ln(x - 1) + \cos(x - 1) = 0$ for $1.3 \leq x \leq 2$
 - $2x \cos 2x - (x - 2)^2 = 0$ for $2 \leq x \leq 3$ and $3 \leq x \leq 4$
 - $(x - 2)^2 - \ln x = 0$ for $1 \leq x \leq 2$ and $e \leq x \leq 4$
 - $e^x - 3x^2 = 0$ for $0 \leq x \leq 1$ and $3 \leq x \leq 5$
 - $\sin x - e^{-x} = 0$ for $0 \leq x \leq 1$, $3 \leq x \leq 4$ and $6 \leq x \leq 7$

26. The accumulated value of a savings account based on regular periodic payments can be determined from the *annuity due equation*,

$$A = \frac{P}{i}[(1 + i)^n - 1].$$

In this equation, A is the amount in the account, P is the amount regularly deposited, and i is the rate of interest per period for the n deposit periods. An engineer would like to have a savings account valued at \$750,000 upon retirement in 20 years and can afford to put \$1500 per month toward this goal. What is the minimal interest rate at which this amount can be invested, assuming that the interest is compounded monthly?

27. Problems involving the amount of money required to pay off a mortgage over a fixed period of time involve the formula

$$A = \frac{P}{i}[1 - (1 + i)^{-n}],$$

known as an *ordinary annuity equation*. In this equation, A is the amount of the mortgage, P is the amount of each payment, and i is the interest rate per period for the n payment periods. Suppose that a 30-year home mortgage in the amount of \$135,000 is needed and that the borrower can afford house payments of at most \$1000 per month. What is the maximal interest rate the borrower can afford to pay?

28. A drug administered to a patient produces a concentration in the blood stream given by $c(t) = Ate^{-t/3}$ milligrams per milliliter, t hours after A units have been injected. The maximum safe concentration is 1 mg/mL.
- What amount should be injected to reach this maximum safe concentration, and when does this maximum occur?
 - An additional amount of this drug is to be administered to the patient after the concentration falls to 0.25 mg/mL. Determine, to the nearest minute, when this second injection should be given.
 - Assume that the concentration from consecutive injections is additive and that 75% of the amount originally injected is administered in the second injection. When is it time for the third injection?

20. The equation $x^2 - 10 \cos x = 0$ has two solutions, ± 1.3793646 . Use Newton's method to approximate the solutions to within 10^{-5} with the following values of p_0 .

- a. $p_0 = -100$ b. $p_0 = -50$ c. $p_0 = -25$
d. $p_0 = 25$ e. $p_0 = 50$ f. $p_0 = 100$

2. Find approximations to within 10^{-5} to all the zeros of each of the following polynomials by first finding the real zeros using Newton's method and then reducing to polynomials of lower degree to determine any complex zeros.

- a. $f(x) = x^4 + 5x^3 - 9x^2 - 85x - 136$
b. $f(x) = x^4 - 2x^3 - 12x^2 + 16x - 40$
c. $f(x) = x^4 + x^3 + 3x^2 + 2x + 2$
d. $f(x) = x^5 + 11x^4 - 21x^3 - 10x^2 - 21x - 5$
e. $f(x) = 16x^4 + 88x^3 + 159x^2 + 76x - 240$

Use Crout factorization for tridiagonal systems to solve the following linear systems.

- a. $2x_1 + x_2 = 3,$
 $x_1 + 2x_2 + x_3 = -2,$
 $2x_2 + 3x_3 = 0.$
- b. $2x_1 - x_2 = 5,$
 $-x_1 + 3x_2 + x_3 = 4,$
 $x_2 + 4x_3 = 0.$
- c. $2x_1 - x_2 = 3,$
 $x_1 + 2x_2 - x_3 = 4,$
 $x_2 - 2x_3 + x_4 = 0,$
 $x_3 + 2x_4 = 6.$
- d. $2x_1 - x_2 = 1,$
 $x_1 + 2x_2 - x_3 = 2,$
 $2x_2 + 4x_3 - x_4 = -1,$
 $2x_4 - x_5 = -2,$
 $x_4 + 2x_5 = -1.$

Solve system of equations by Jacobi's Method and by Gauss-Seidel Method

- a. $4x_1 + x_2 - x_3 = 5,$
 $-x_1 + 3x_2 + x_3 = -4,$
 $2x_1 + 2x_2 + 5x_3 = 1.$
- b. $-2x_1 + x_2 + \frac{1}{2}x_3 = 4,$
 $x_1 - 2x_2 - \frac{1}{2}x_3 = -4,$
 $x_2 + 2x_3 = 0.$
- c. $4x_1 + x_2 - x_3 + x_4 = -2,$
 $x_1 + 4x_2 - x_3 - x_4 = -1,$
 $-x_1 - x_2 + 5x_3 + x_4 = 0,$
 $x_1 - x_2 + x_3 + 3x_4 = 1.$
- d. $4x_1 - x_2 - x_4 = 0,$
 $-x_1 + 4x_2 - x_3 - x_5 = 5,$
 $-x_2 + 4x_3 - x_6 = 0,$
 $-x_1 + 4x_4 - x_5 = 6,$
 $-x_2 - x_4 + 4x_5 - x_6 = -2,$
 $-x_3 - x_5 + 4x_6 = 6.$